Zooplankton Hierarchical Classification

CNN-RNN Sequence of Tokens Approach

Philip Loewen - University of Toronto, Department of Statistical Sciences

The Problem

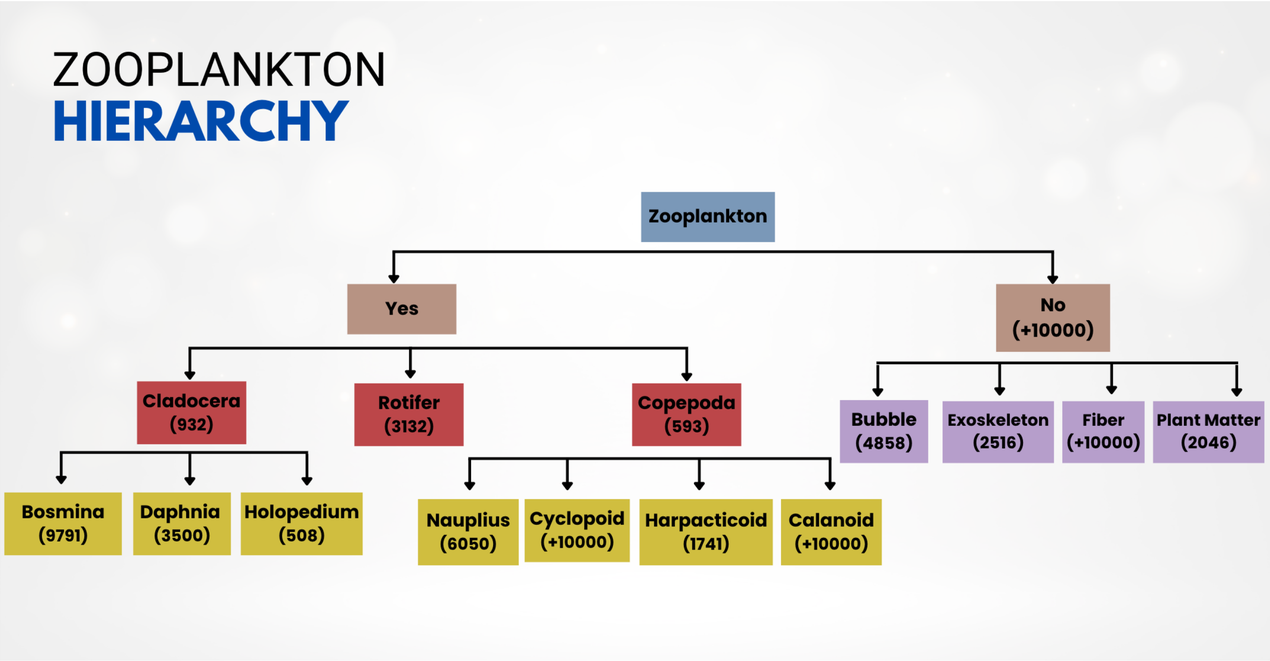
What are we classifying?

12 classes of images with 8 species of zooplankton from microscopy images spanning cladocera, rotifers, copepods.

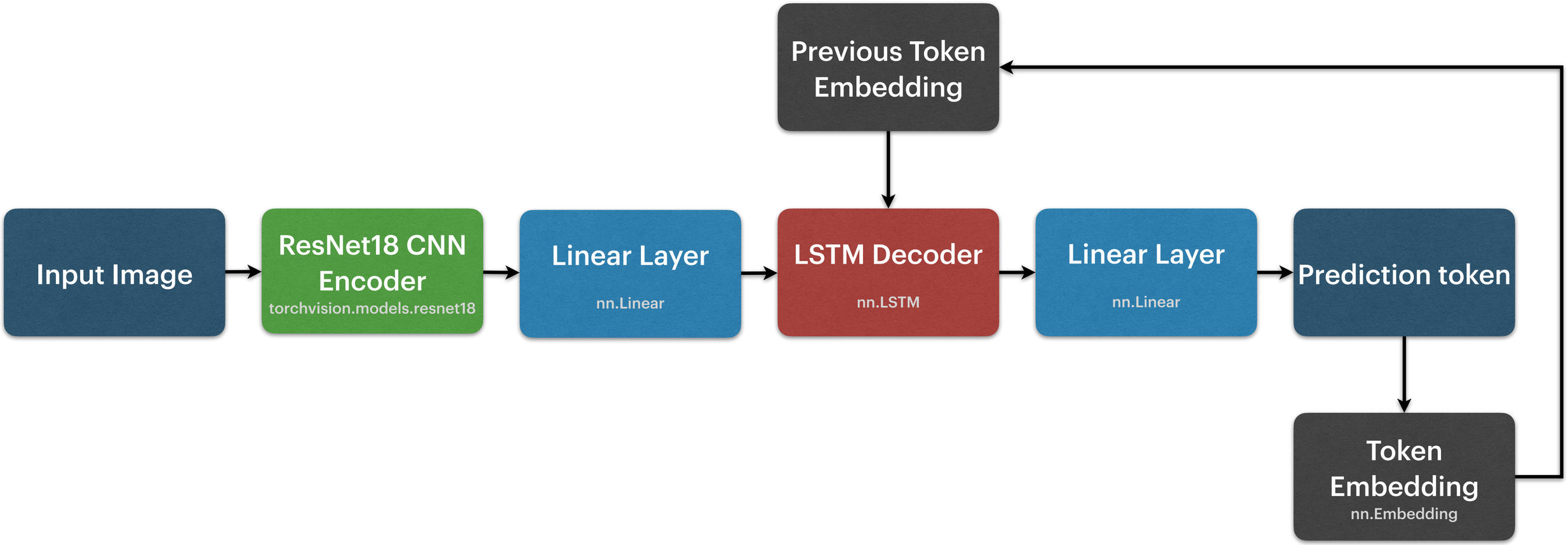
Why is it difficult?

Classes share a natural taxonomic hierarchy which classifiers ignore.

This structure improves model interpretability and help us understand why errors occur.



Model Architecture

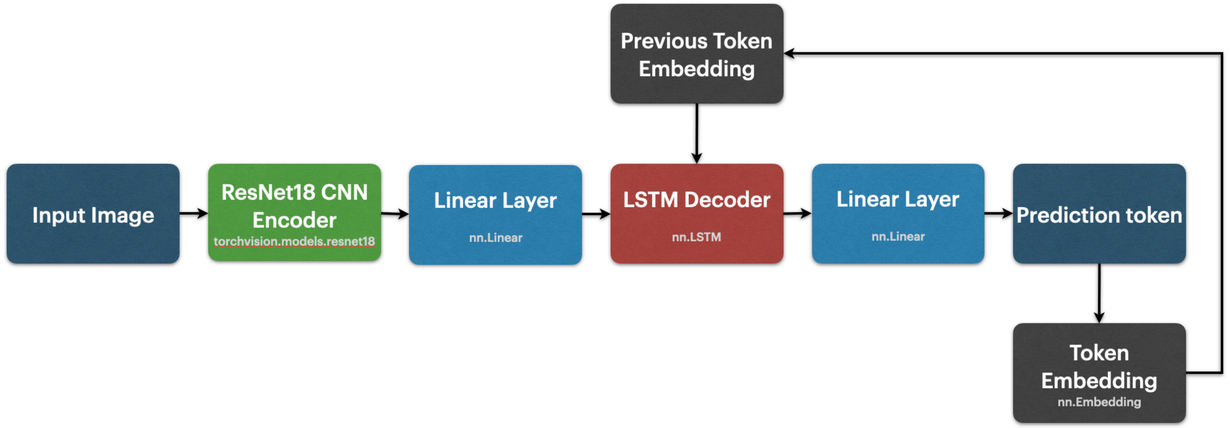


Example output: <START> → Root → Yes → Copepoda → Calanoid → <END>

Model Architecture

Why use this architecture ?

- **Follows naturally** from the problem



- Hierarchy can be **easily changed**

- Reuses LSTM module **reducing parameter count**

Changing The Hierarchy

```
TAXONOMY = {
    "Root": ["Yes", "No"],
    "Yes": ["Cladocera", "Rotifer_Group", "Copepoda"],
    "No": ["Bubble_Group", "Exoskeleton_Group", "Fiber", "Plant_Matter_Group"],
    "Cladocera": ["Bosminidae", "Daphnia"],
    "Copepoda": ["Calanoid", "Cyclopoid", "Harpacticoid", "Nauplius_Copepod"],
    "Rotifer_Group": ["Rotifer"],
    "Bubble_Group": ["Bubbles"],
    "Exoskeleton_Group": ["Exoskeleton"],
    "Fiber": ["Fiber_Hairlike", "Fiber_Squiggly"],
    "Plant_Matter_Group": ["Plant_Matter"],
}
```

Training Loop

- 60/20/20 **Training/Val/Test**
- **Adam Optimizer** with **Gradient Clipping** max norm 1.0
- **Cross Entropy Loss** computed on flattened sequences jointly
- **Teacher Forcing Mode** feeds ground truth predictions as decoder outputs, making the model predict the next token
- **Dataset Augmentation** by rotating and flipping images for smaller classes

Parameter	Value
Embedding Size	512
Hidden Dimension Size	512
Learning Rate	0.0001
Max Epochs	50
Early Stopping Threshold	0.003
Early Stopping Patience	10 epochs

Tuned Hyperparameters

Loss Computation

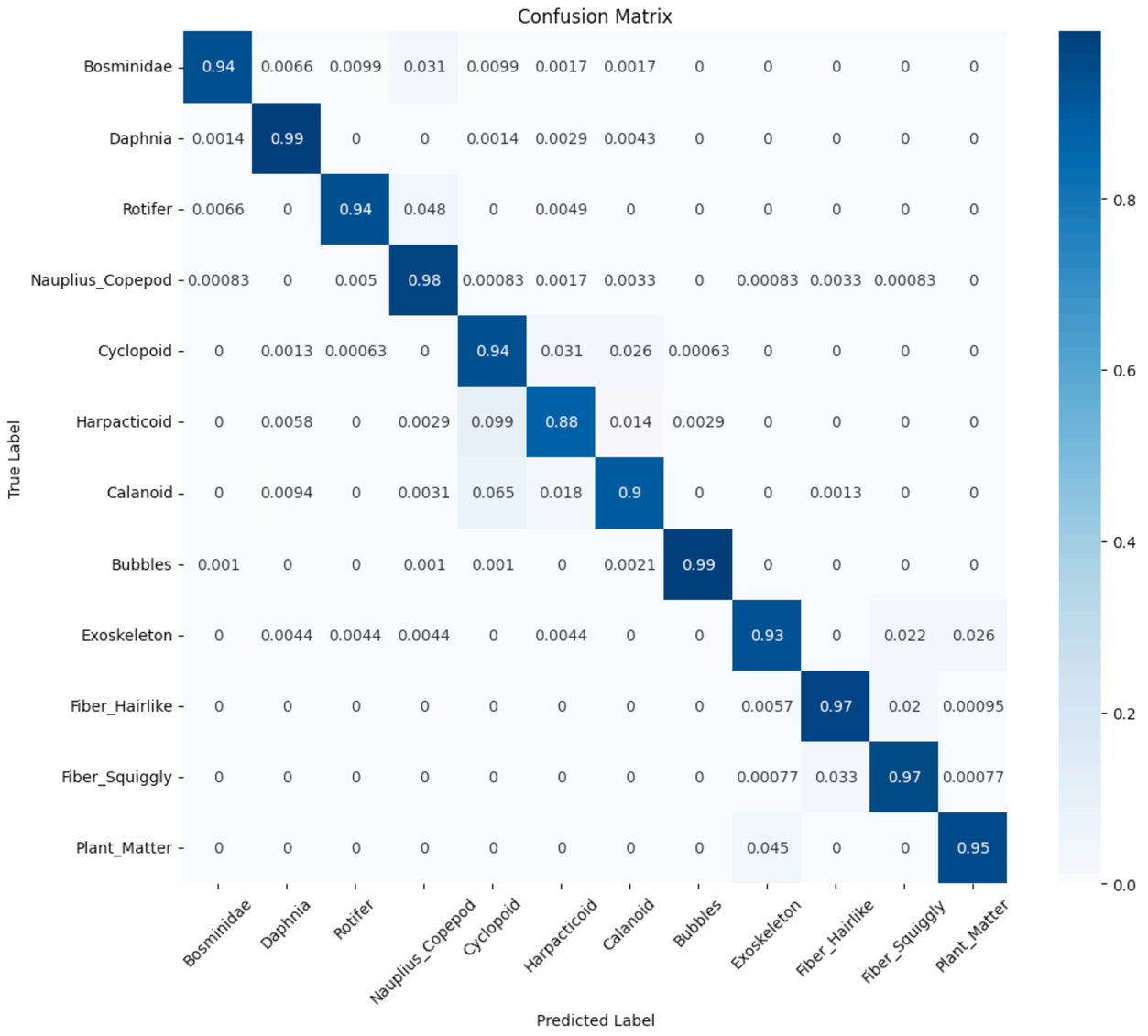
Each component of the prediction is treated independently of each other

	Raw Form	**Flattened Form**
Prediction	[[“Yes”, “Cladocera”, “Bosmina”], [“No”, “Fiber”, “Fiber Squiggly]]	[“Yes”, “Cladocera”, “Bosmina”, “No”, “Fiber”, “Fiber Squiggly”]
Truth	[[“Yes”, “Copepoda”, “Nauplius”], [“No”, “Fiber”, “Fiber Hairlike”]]	[“Yes”, “Copepoda”, “Nauplius”, “No”, “Fiber”, “Fiber Hairlike”]

Classification Results

Leaf Level

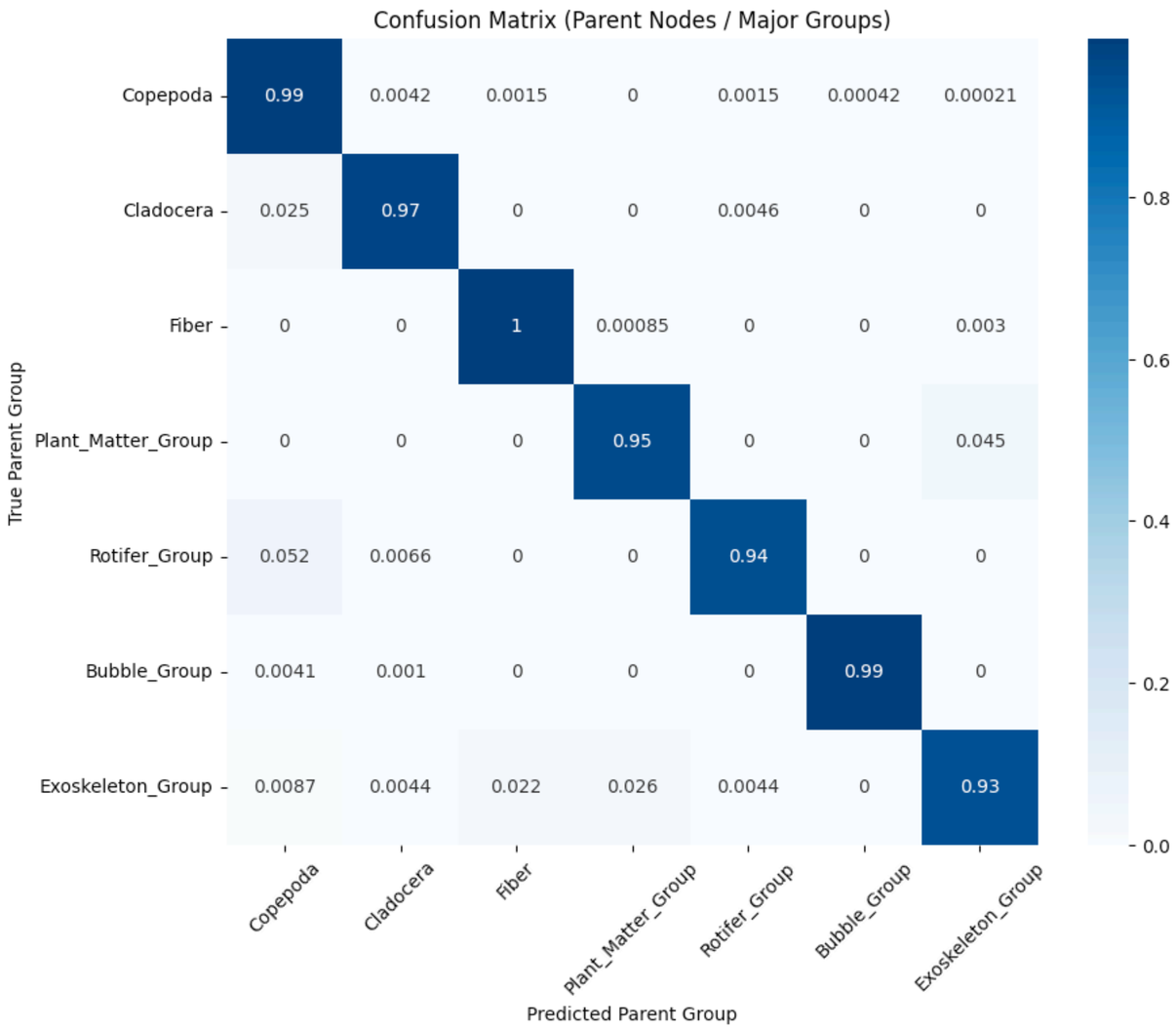
- **95% Total Accuracy & F1 Score**
- **Best Predictions** on Daphnia (Cladocera), Nauplius (Copepod), Bubbles
- **Worst Predictions** on Harpacticoid (Copepod)



Classification Results

Parent Level

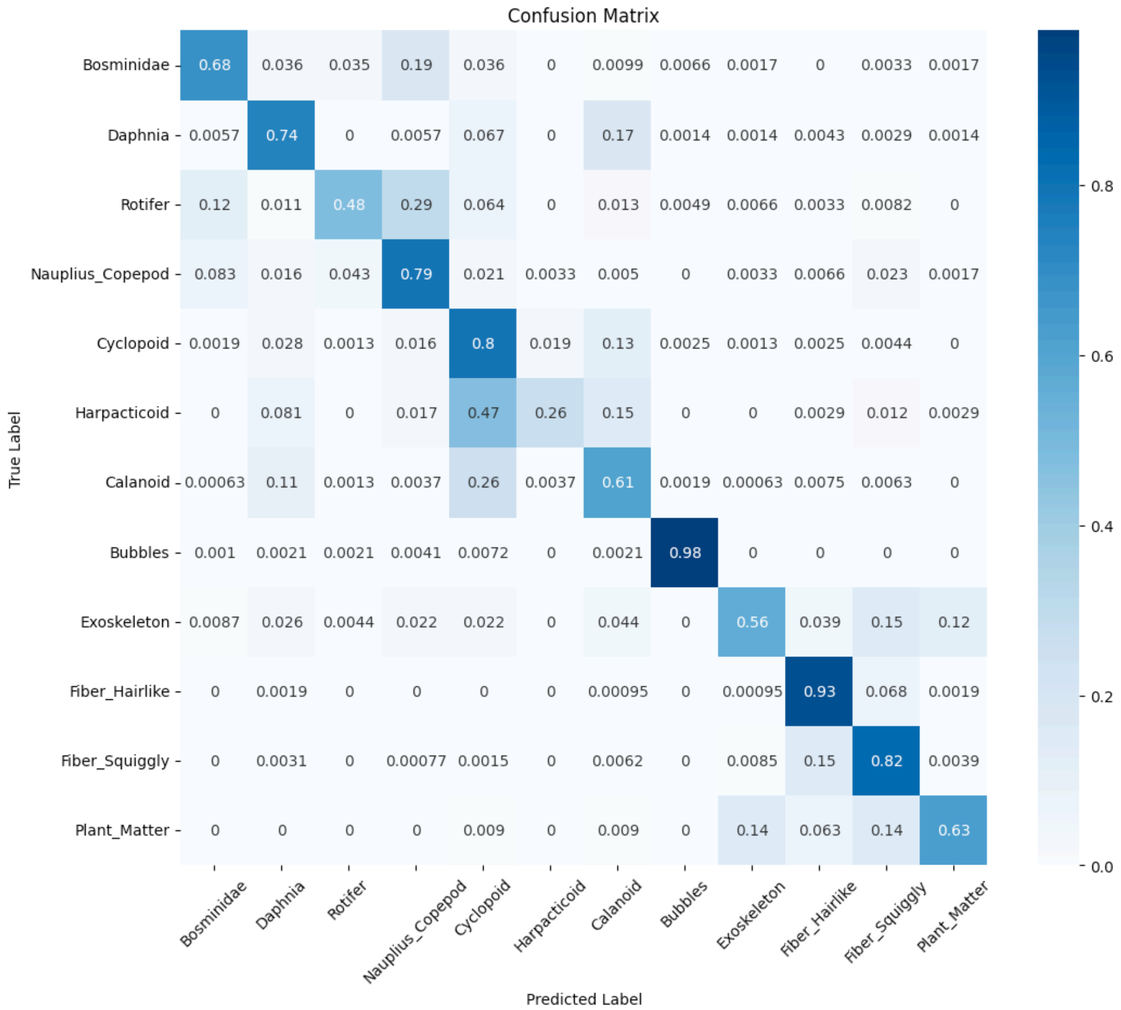
- **99% Total Accuracy & F1 Score**
- **Best Predictions** on Fiber, Copepoda, Bubbles
- **Worst Predictions** on Rotifer, Exoskeleton



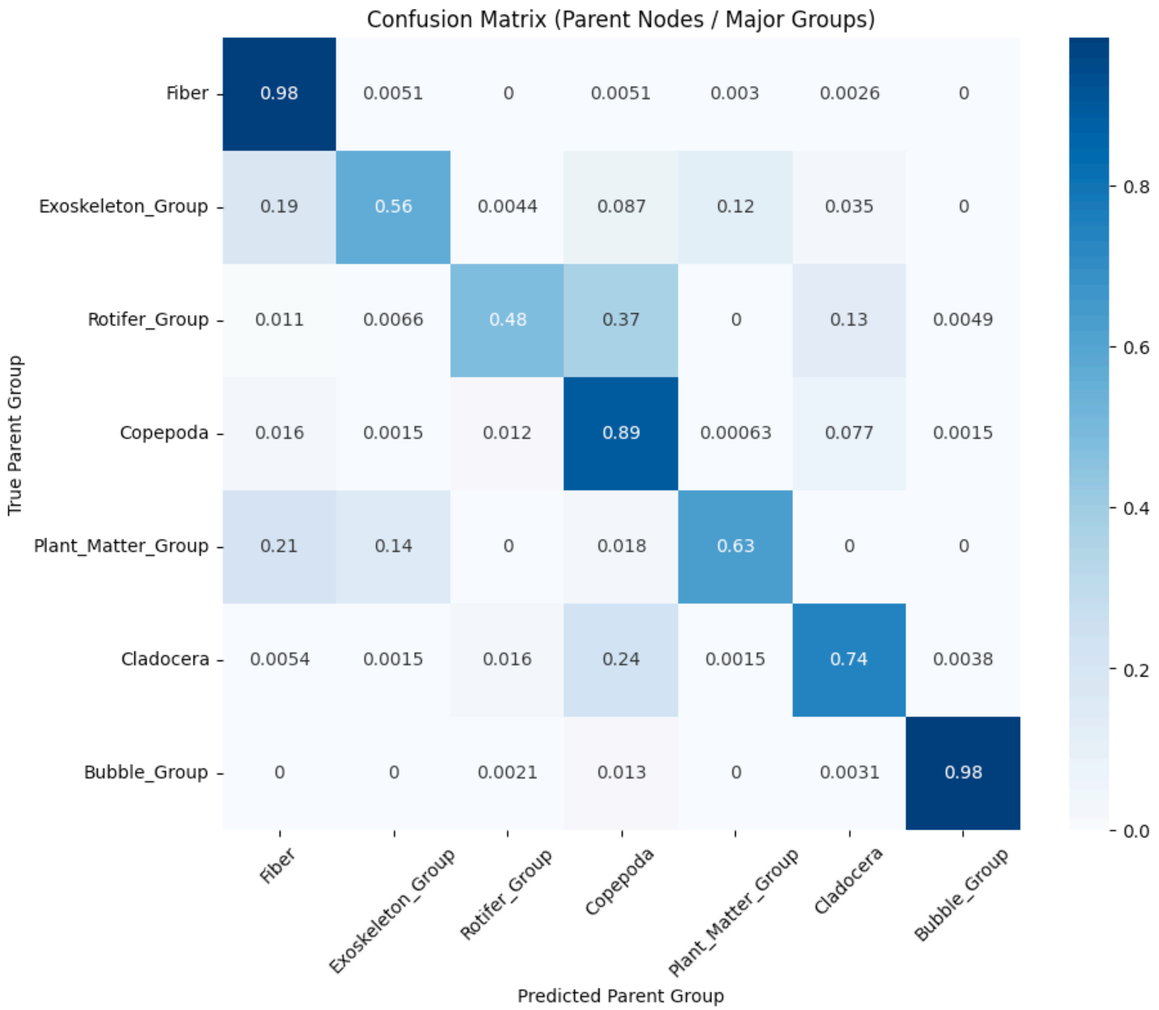
Thank You

Appendix

Frozen ResNet Results



Leaf Level: 75% Accuracy



Parent Level: 87% Accuracy